

OpenModelica And The Wonderful World Of SIMULATION

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Real-life superhero Elon Musk recently speculated whether we all might be living in some kind of Wachowski siblings' kind of matrix, without even knowing. I personally think that it is not possible, although if it were the case, may be I am programmed to think that it was not. Enough with the what ifs and what nots, let us talk about the hot 'S' word, simulation.

Why simulate

All systems we see around us are tested and looked at extensively via some sort of simulation. Engineers and designers have a plethora of tools to see exactly how the piston they designed would behave in an enclosed cylinder casing at temperatures ranging from 100 degrees to 200 degrees. You name the condition and there are simulators that can easily incorporate that aspect into the test.

It is amazing how different systems can be made to interact with each other as well. Just like the above example, scientists can see how a nut will behave in zero gravity, while exposed to a bunch of cosmic rays. If need be, you can simulate the conditions that would make you the next Spider Man. Let us calm down though, there is a lot to learn before we get there.

In this edition of *Electronics For You Plus*, we introduce you to one of the pioneers of the present simulation scene and then delve deeper into a spin-off that caters to the education sector. This month, we look at OpenModelica, the free

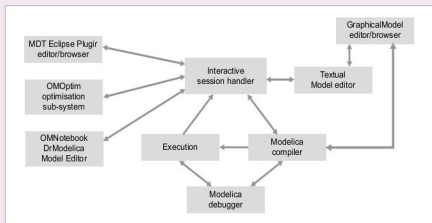


Fig. 1: Architecture of OpenModelica Environment

open source environment based on Modelica modelling language.

Modelica is a tool to model, simulate, optimise and analyse complex dynamic systems. It is run by the good folks at Open Source Modelica Consortium, a non-profit, non-government organisation committed to keeping this software alive and well.

What Modelica is

Before we dig into the idea of OpenModelica, let us take a look at the mother ship, Modelica. That should help us appreciate the thought that has gone behind the education-heavy OpenModelica. Modelica is an object-oriented, declarative, multi-domain modelling language that can be used to model systems of a variety of kinds. Examples of such systems could be electrical, mechanical, electronic, hydraulic, thermal, control, electric power or process-oriented sub-components. Any system that has components of any of the variety

mentioned above can be simulated using Modelica. Managed by Modelica Association, as of version 3.2.1, Modelica Standard Library contains about 1360 generic model components and 1280 functions in a bevy of domains.

Modelica in its basic form looks a lot like C++ or, for that matter, Java. This can be attributed to the object-oriented programming it employs. The difference, of course, is that it is not a programming language, but rather a modelling language. Compiling Modelica code also brings out differences as it is compiled by first converting the code into objects, to be further processed by the respective simulation engine.

Execution of commands also takes a different approach from C++ code. While the latter lays stress on assignment statements, Modelica incorporates a series of equations that have to be solved to explain the behaviour of the system.